

JEE Main - 2 | JEE 2022 (Gen 1 & 2)

Date: 24/08/2020

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99 (both inclusive).
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

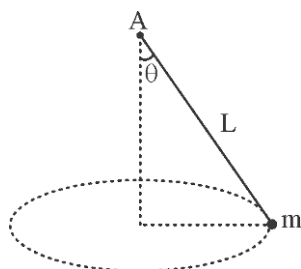
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

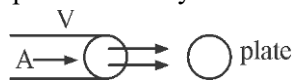
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

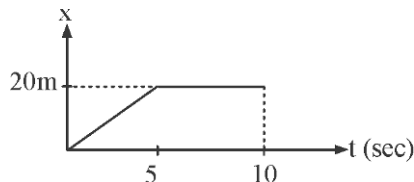
- The acceleration of a particle varies with displacement as $a = 4x$. Find the velocity of the particle at $x = 10\text{m}$. The particle was initially at rest $x = 0\text{m}$ {Assume x to be in metre and a in m/s^2 }
 (A) 10 m/s (B) 20 m/s (C) 30 m/s (D) 5 m/s
- A bob of mass m is connected to a massless string of length L which is hinged at point A as shown in the figure. If the bob is rotated with constant speed in a horizontal circle such that the string makes an angle 30° with the vertical, find the speed of the bob.



- (A) $\left(\frac{Lg}{\sqrt{3}}\right)^{1/2}$ (B) $\left(\frac{2Lg}{\sqrt{3}}\right)^{1/2}$ (C) $\left(\frac{Lg}{2\sqrt{3}}\right)^{1/2}$ (D) $\left(\frac{\sqrt{3}Lg}{2}\right)^{1/2}$
- A liquid of density ρ is flowing with speed V through a pipe of cross-sectional area A . The liquid after coming out of pipe hits the plate and splashes along the surface at right angles to the original motion. Find the force F required to hold the plate stationary.

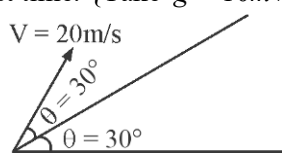


- (A) ρAV^2 (B) $\sqrt{2}\rho AV^2$ (C) $2\rho AV^2$ (D) $3\rho AV^2$
- A hot air balloon is moving up with a constant velocity. A ball is dropped from the balloon at any time. Which of the following observations is correct just after dropping the ball? {A is a stationary observer on the ground and B is a man standing on the balloon}
 (A) Both A and B feel that ball is going up
 (B) Both A and B feel that ball is going down
 (C) A feels that ball is coming down, while B feels that the ball is going up
 (D) A feels that ball is going up while B feel that ball is going down
- The position – time graph of a car starting from rest at $t = 0$ is shown in the figure. Find the speed of the car at time $t = 8$ seconds.

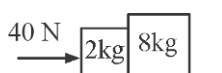


- (A) 4 m/s (B) 8 m/s (C) 2 m/s (D) Zero

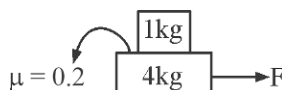
6. If particle is projected with speed 20 m/s at an angle 30° with inclined plane then find the time when it will hit the inclined plane for the first time. {Take $g = 10 \text{ m/s}^2$ }



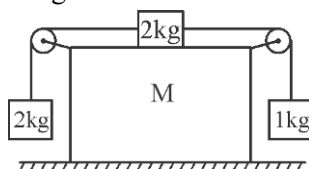
- (A) $\frac{\sqrt{3}}{4} \text{ s}$ (B) $\frac{4}{\sqrt{3}} \text{ s}$ (C) $\frac{2}{\sqrt{3}} \text{ s}$ (D) $\sqrt{3} \text{ s}$
7. Find the velocity of projection if range and maximum height of projectile are 12m and 4m respectively. (g is acceleration due to gravity)
- (A) $5\sqrt{\frac{g}{2}}$ (B) $3\sqrt{\frac{g}{2}}$ (C) $\frac{1}{3}\sqrt{\frac{g}{2}}$ (D) $\frac{1}{5}\sqrt{\frac{g}{2}}$
8. A ball is projected with velocity u and angle θ with the ground (horizontal). Considering point of projection as origin, the displacement of particle in horizontal axis- x and vertical axis- y is given as $x = 6t$, $y = 8t - 5t^2$. Find the initial velocity of projection (u). {Take $g = 10 \text{ m/s}^2$.}
- (A) 6 m/s (B) 8 m/s (C) 10 m/s (D) 14 m/s
9. Two stones are projected with same speed such that they have different angles of projection but same range. If the angle of projection of one is $\frac{\pi}{3}$ and its maximum height is h_1 then maximum height of other will be:
- (A) $3h_1$ (B) $h_1/3$ (C) $2h_1$ (D) $h_1/2$
10. Two blocks of 2kg and 8kg are kept on a smooth horizontal surface in contact with each other. If a force of 40N is applied to a block of 2kg as shown in the figure, find the Normal force applied by 8kg block on 2kg block.



- (A) 40 N (B) 32 N (C) 20 N (D) 8 N
11. If coefficient of friction between blocks is $\mu = 0.2$, then find the max force F which can be applied so that there is no slipping between them. The surface below 4 kg block is smooth.

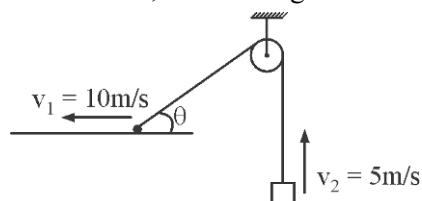


- (A) 20 N (B) 10 N (C) 8 N (D) 2 N
12. A smooth block of mass M with 2 pulleys attached to it is fixed on the ground. A 2kg block is kept on it and is attached to two other blocks through massless strings as shown in the figure. Find the tension in the string connecting 2kg block and 1 kg block.



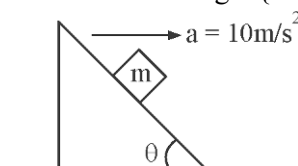
- (A) 20 N (B) 10 N (C) 12 N (D) 6 N

13. A massless string connects two particles as shown in the figure. If at any given instant, velocities of both the particles are 10 m/s and 5 m/s as shown, find the angle θ at that instant.



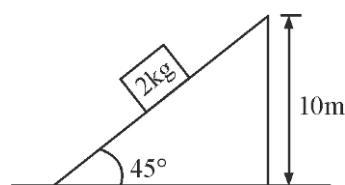
- (A) 30° (B) 60° (C) 45° (D) None of these

14. Wedge is moved along horizontal with acceleration $a = 10 \text{ m/s}^2$ such that the block doesn't move with respect to wedge. Find the angle of inclination of wedge. {Assume all surfaces to be smooth}



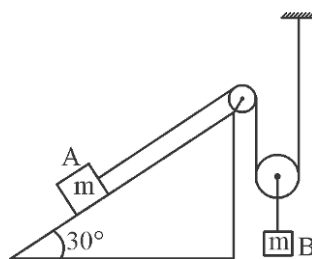
- (A) 30° (B) 60° (C) 45° (D) 15°

15. If the wedge is fixed and there is friction between the block and wedge, then find the time required by block to reach the ground if it is left from the top of incline. {Take $g = 10 \text{ m/s}^2$ & $\mu_k = \frac{1}{3}$ }



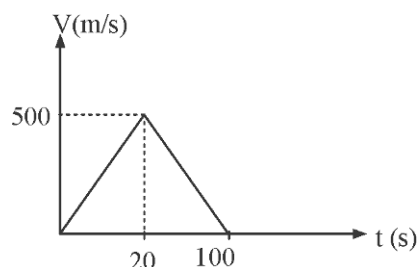
- (A) 3 sec (B) $\sqrt{3}$ sec (C) 6 sec (D) $\sqrt{6}$ sec

16. Two blocks A and B of same mass m are connected through massless strings as shown in the figure. The wedge is smooth and fixed. Find acceleration of blocks A and B respectively when the system is released.



- (A) $\frac{g}{3}, \frac{g}{6}$ (B) $\frac{g}{6}, \frac{g}{3}$ (C) $\frac{g}{2}, \frac{g}{2}$ (D) 0, 0

17. The $v-t$ graph of a racing car from one stop to another stop is shown. Find the distance between the two stops.



- (A) 1 km (B) 5 km (C) 20 km (D) 25 km

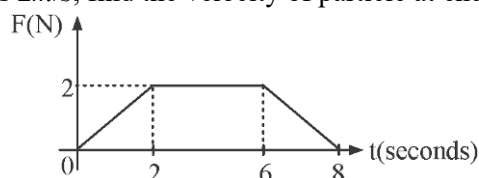
18. If a ball is thrown upward with speed V from the top of a building and it reaches the ground with speed of $4V$, what is the height of building? { g = acceleration due to gravity}
- (A) $\frac{15V^2}{2g}$ (B) $\frac{15V^2}{g}$ (C) $\frac{9V^2}{2g}$ (D) $\frac{9V^2}{g}$
19. A particle is moving in a circle of radius $1m$. The velocity of the particle varies with time as $v = 2t \text{ m/s}$. What is the acceleration of the particle at $t = 2\text{seconds}$?
- (A) $2\sqrt{65} \text{ m/s}^2$ (B) $\sqrt{65} \text{ m/s}^2$ (C) $5\sqrt{13} \text{ m/s}^2$ (D) None of these
20. If velocity of a particle having mass 2kg varies with time as $4t^2$ then find the net force experienced by it at $t = 4 \text{ sec}$.
- (A) 16 N (B) 32 N (C) 64 N (D) 128 N
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SPACE FOR ROUGH WORK

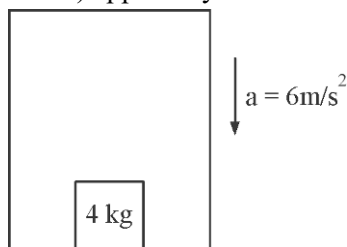
SECTION-2

This Section contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99 (both inclusive).

1. A man standing in a lift throws a ball with a speed of 24 m/s in upward direction relative to lift and lift is moving upwards with an acceleration $a = 2 \text{ m/s}^2$. Find the time t when ball comes back to man in seconds. {Take $g = 10 \text{ m/s}^2$ } {Assume the ball does not hit the roof of lift}.
2. A particle of mass 2 kg is acted upon by a force which varies with time as shown in the figure. If the velocity of particle at $t = 0$ is 2 m/s , find the velocity of particle at end of 8 seconds. {in m/s .}



3. A lift is moving down with an acceleration of 6 m/s^2 . If a block of mass 4 kg is kept on the floor of the lift, find the normal reaction (in Newton) applied by block on the floor of the lift. {Take $g = 10 \text{ m/s}^2$ }



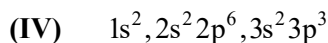
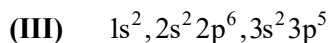
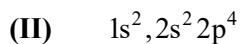
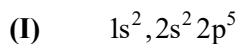
4. From top of a tower of height 80 m two stones are projected horizontally with 20 m/s & 40 m/s in opposite directions. The distance between these two stones will be $10x$ meters when they reach ground. Find the value of x . {Take $g = 10 \text{ m/s}^2$ }
5. A car of mass 100 kg moves on a horizontal circular road of radius 20 m . The coefficient of static friction between tyres and road is $\frac{1}{2}$. Find the maximum speed {in m/s } with which car can move in the circle without slipping. {Take $g = 10 \text{ m/s}^2$ }

SPACE FOR ROUGH WORK

PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

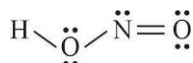
1. The electronic configuration of four elements (I), (II), (III) and (IV) are given below:



Which element has highest value of 1st Electron gain enthalpy (magnitude)?

- (A) (I) (B) (II) (C) (III) (D) (IV)

2. In the Lewis structure of nitrous acid:



What is the formal charge on nitrogen?

- (A) -1 (B) 0 (C) +1 (D) +3

3. A polar molecule AB has dipole moment 3.2 D (Debye) while the bond length is 1.6 Å. Find the percentage ionic character in the molecule.

- (A) 31% (B) 41.6% (C) 39.6% (D) None of these

4. The energy of the first orbit of hydrogen atom is -13.6 eV. If the fourth ionization enthalpy of beryllium atom is expressed as $Y \times 10^{-19} \text{ J}$, what is the value of 'Y'?

- (A) 0.34816 (B) 3.4816 (C) 34.816 (D) 348.16

5. When the surface of a piece of potassium is irradiated with light of wavelength 5000 Å, the kinetic energy of the ejected electron is $0.32 \times 10^{-19} \text{ J}$. When the wavelength of light is changed to 4000 Å, the kinetic energy of the ejected electron changes to $0.96 \times 10^{-19} \text{ J}$. In this problem, assume that YOU DO NOT KNOW the value of Planck's constant. Using this information calculate the value of Planck's constant (in J-sec) ($c = 3 \times 10^8 \text{ m/s}$)

- (A) 6.63×10^{-34} (B) 3.60×10^{-34} (C) 2.14×10^{-34} (D) 4.27×10^{-34}

6. In which of the following species $p\pi - d\pi$ bond is present but $p\pi - p\pi$ bond is absent?

- (A) SiH_4 (B) CS_2 (C) SO_2 (D) SO_2Cl_2

7. Which of the following sets have correctly matched each molecule or ion and its shape?

	Tetrahedral	Trigonal pyramidal	T-shaped	Square planar
(A)	CH_4	BCl_3	NO_3^-	SO_4^{2-}
(B)	SO_4^{2-}	NF_3	ICl_3	XeF_4
(C)	CH_4	NO_3^-	GaI_3	SnCl_4
(D)	CCl_4	PF_3	ICl_3	SF_4

8. During an experiment, an α -particle and a proton are accelerated by same potential difference, their de-Broglie wavelength ratio will be: (Take mass of proton = mass of neutron)

- (A) 1 : 2 (B) 1 : 4 (C) $1 : 2\sqrt{2}$ (D) $1 : \sqrt{2}$

9. The correct order of the increasing ionic character is:

- (A) $\text{BeCl}_2 < \text{MgCl}_2 < \text{CaCl}_2 < \text{BaCl}_2$ (B) $\text{BeCl}_2 < \text{MgCl}_2 < \text{BaCl}_2 < \text{CaCl}_2$
 (C) $\text{BeCl}_2 < \text{BaCl}_2 < \text{MgCl}_2 < \text{CaCl}_2$ (D) $\text{BaCl}_2 < \text{MgCl}_2 < \text{CaCl}_2 < \text{BeCl}_2$

10. The successive ionization energies (in kJ/mol) for an element are shown below.

IE ₁	IE ₂	IE ₃	IE ₄	IE ₅
577	1820	2740	11600	14800

What is the electron configuration of this element?

- (A) $1s^2, 2s^2 2p^6, 3s^1$ (B) $1s^2, 2s^2 2p^6, 3s^2 3p^1$
 (C) $1s^2, 2s^2 2p^6, 3s^2 3p^3$ (D) $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^3$
11. For an electron, with $n = 3$ has only one radial node. The orbital angular momentum of the electron will be:
 (A) 0 (B) $\sqrt{6} \frac{h}{2\pi}$ (C) $\sqrt{2} \frac{h}{2\pi}$ (D) $\sqrt{12} \left(\frac{h}{2\pi} \right)$
12. The shortest wavelength of He^+ ion in Balmer series is x , then longest wavelength in the Paschen series of Li^{+2} is:
 (A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$
13. Which among the following orbital diagram violates Pauli's exclusion principle, Hund's rule as well as Aufbau's rule?
 (A)

↑	↑↓	↓	
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 (B)

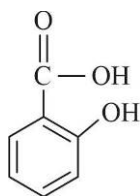
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 (C)

↑↑	↑	↑	↓
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 (D)

↑	↑	↓↓	
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14. If number of unpaired electrons in $\text{Mn}^{+7} = a$
 Number of d-subshell electrons in $\text{Cr} = b$
 Find the value of $(a + b)$
 (Atomic no. of $\text{Mn} = 25$, $\text{Cr} = 24$)
 (A) 12 (B) 9 (C) 11 (D) 5
15. The size of an atomic nucleus is 10^{-14} m . What would be the uncertainty in speed (ms^{-1}) of an electron if it were to exist inside the nucleus? [$h = 6.63 \times 10^{-34} \text{ J-sec}$, $m_e = 9.1 \times 10^{-31} \text{ Kg}$]
 (A) 5.27×10^{11} (B) 5.8×10^8 (C) 5.8×10^{10} (D) 5.8×10^9
16. For H-atom, as the orbit number increases, the distance between two successive orbits ($r_1 =$ radius of first orbit)
 (A) Increases by $2r_1$
 (B) Increases by $(2n+1)r_1$, where n is higher orbit
 (C) Increases by $(2n+1)r_1$, where n is lower orbit
 (D) Remains constant
17. $\text{K}^+, \text{Cl}^-, \text{Ca}^{2+}, \text{S}^{2-}$ ions are isoelectronic. The decreasing order of their size is:
 (A) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$ (B) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$
 (C) $\text{K}^+ > \text{Cl}^- > \text{Ca}^{2+} > \text{S}^{2-}$ (D) $\text{Cl}^- > \text{S}^{2-} > \text{Ca}^{2+} > \text{K}^+$
18. How many resonating structures can be written for the nitrate ion, NO_3^- ?
 (A) 1 (B) 2 (C) 3 (D) 4



19. How many σ - and π -bonds are there in (salicylic acid)?

(A) $12\sigma, 4\pi$

(B) $16\sigma, 4\pi$

(C) $18\sigma, 2\pi$

(D) $16\sigma, 2\pi$

20. What will the total number of lone pairs of electrons be in I_3^- ion?

(A) 3

(B) 4

(C) 9

(D) 8

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is an **integer** ranging from 0 to 99 (both inclusive).

1. Count the total number of unshared electrons in $I_3^- = x$
 Count the total number of $p\pi - d\pi$ bonds in $PO_4^{3-} = y$
 Count the total number of lone pairs in $XeF_4 = z$
 Hence, find the value of $x + y + z$
2. How many of the following species contain one or more lone pair(s) on central atom, that is present in an orbital having contribution from d-orbital.
 SOF_4 $SeOCl_2$ SeF_3^+
 XeF_5^+ BrF_6^- TeF_5^- ICl_4^-
3. First and second ionization energies of magnesium atom are 7.646 and 15.035 eV respectively. The amount of energy in kJ needed to convert all the atoms of magnesium into Mg^{2+} ions present in 12 mg of magnesium vapours is: [Given : $1 \text{ eV} = 100 \text{ kJ mol}^{-1}$] (Report your answer to the nearest whole number) [Molar mass of Mg = 24g/mol]
4. Number of possible spectral lines which may be emitted in brackett series in H atom, if electron present in 9^{th} excited level returns to ground level, are
5. If $3e^-$ can be accommodated in each orbital then how many elements will be present in the 7^{th} period of periodic table?

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If roots of the equation $z^8 = 1$ are $1, z_1, z_2, z_3, z_4, z_5, z_6, z_7$ then the value of $\sum_{k=1}^7 z_k^7$ is :
 (A) 0 (B) -1 (C) 6 (D) 5
- The value of $\sum_{n=0}^{100} i^{(n+1)!}$ is equal to : [Note : where $i = \sqrt{-1}$]
 (A) -1 (B) i (C) $2i + 95$ (D) $96 + i$
- The sum to n terms of the series $12 + 104 + 1006 + \dots$ is :
 (A) $\frac{1}{9}(10^n - 1) + n^2 + n$ (B) $\frac{1}{9}(10^n - 1) + 2n$
 (C) $\frac{10}{9}(10^n - 1) + n^2 + n$ (D) $\frac{10}{9}(10^n - 1) + 2n$
- Let a, b are roots of $z^2 + z + 1 = 0$, then:
 (A) $a^2 + b^2 = 4$ (B) $(a - b)^2 = 3$ (C) $a^3 + b^3 = 2$ (D) $a^4 + b^4 = 1$
- Let p_r denote the sum of first r terms of A.P. whose first term is $r + 1$ and common difference is $2r + 2$, then $p_1 + p_2 + p_3 \dots p_n$ is:
 (A) $\frac{n(n+1)(n+2)^2}{6}$ (B) $\frac{n(n+1)(n+3)(3n-1)}{12}$
 (C) $\frac{n(n+1)(n+2)(3n+1)}{12}$ (D) None of these
- Let $S_n = \sum_{k=1}^n (-1)^{\frac{k(k+1)}{2}} k^2$, then S_{19} is:
 (A) 400 (B) 380 (C) 420 (D) None of these
- If z is complex lying in 2nd quadrant and $|\operatorname{Re}(z)| > |\operatorname{Im}(z)|$ then complex $\frac{\bar{z}}{z}$ lie in:
 (A) 1st Quadrant (B) 2nd Quadrant (C) 3rd Quadrant (D) 4th Quadrant
- If $\sum_{r=1}^n T_r = (n^2 + 3n)^2 + 2 \cdot (n^2 + 3n)$, then $\sum_{n=1}^{\infty} \frac{1}{T_n}$ is _____.
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

9. If $z = \cos \alpha + i \sin \alpha$, then polar form of $z^2 - \bar{z}$ is: $\left(0 < \alpha < \frac{2\pi}{3}\right)$
- (A) $2 \left[\sin \frac{3\alpha}{2} \right] e^{i(\alpha/2)}$ (B) $2 \left[\sin \frac{3\alpha}{2} \right] e^{i\left(\frac{\pi}{2} + \frac{\alpha}{2}\right)}$
 (C) $-2 \left[\sin \frac{3\alpha}{2} \right] e^{i\left(-\frac{\pi}{2} + \alpha\right)}$ (D) $-2 \left[\sin \frac{3\alpha}{2} \right] e^{i\left(\frac{\pi}{2} + \frac{\alpha}{2}\right)}$
10. Let $z \in \mathbb{C}$ with $\operatorname{Re}(z) = 30$ and it satisfies $\frac{2z-n}{2z+n} = 4i - 1$ for some integer n . Then:
 (A) $n = -45$ and $\operatorname{Im}(z) = -3$ (B) $n = 45$ and $\operatorname{Im}(z) = -5$
 (C) $n = -75$ and $\operatorname{Im}(z) = -15$ (D) $n = -75$ and $\operatorname{Im}(z) = -5$
11. For any complex number $t = p + iq$ where $p, q \in \mathbb{R}$. If $t = \cos\left(\frac{2\pi}{9}\right) + i \sin\left(\frac{2\pi}{9}\right)$, then $|t + 2t^2 + 3t^3 + \dots + 9t^9|^{-1}$ equals:
 (A) $\frac{1}{9} \sin 40^\circ$ (B) $\frac{2}{9} \sin 20^\circ$ (C) $\frac{2}{9} \cos 40^\circ$ (D) $\frac{9}{2} \cos 20^\circ$
12. The maximum value $|z|$ when $\left|z + \frac{2}{z}\right| = 1$ is:
 (A) $\sqrt{3} - 1$ (B) $\sqrt{3} + 1$ (C) 2 (D) 1
13. If $1, \alpha_1, \alpha_2, \dots, \alpha_{10}$ be roots of equation $x^{11} - 1 = 0$ and ω be an imaginary cube root of unity, then $\frac{(\omega^2 - \alpha_1)(\omega^2 - \alpha_2) \dots (\omega^2 - \alpha_{10})}{(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_{10})}$:
 (A) ω (B) $-\omega$ (C) 1 (D) ω^2
14. The sum of the n terms of series $1 + 2^2 + 3 + 4^2 + 5 + 6^2 + \dots$ is $\frac{n}{12}(n+4)(2n+1)$, when n is even. When n is odd the sum is :
 (A) $\frac{(n-1)(n+3)(2n-1)}{12} + n$ (B) $\frac{(n-1)}{12}(n+3)(2n-1) + n^2$
 (C) $\frac{(n-1)}{12}(n+3)(2n+1)$ (D) $\frac{(n-1)}{12}(n+4)(2n+1) + n$
15. Let a, b, c are three arithmetic means between two numbers such that $a + b + c = 30$ and p, q, r be harmonic means between same numbers such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{5}{6}$, then the numbers are :
 (A) 3, 10 (B) 3, 12 (C) 18, 2 (D) 9, 1
16. The positive integers are written in a triangular array as shown
- | |
|----------|
| 1 |
| 2 3 |
| 4 5 6 |
| 7 8 9 10 |
| ----- |
- then row in which the number 100 will be, is :
 (A) 20 (B) 14 (C) 15 (D) 22

17. Let $z = \frac{(2\sqrt{3} + 2i)^8}{(1-i)^6} + \frac{(1+i)^6}{(2\sqrt{3} - 2i)^8}$ where $i = -1$, then :
- (A) $|z| = 2^{13} + \frac{1}{2^{13}}$ and $\text{amp } z = \frac{5\pi}{6}$ (B) $|z| = 2^{12} + \frac{1}{2^{12}}$ and $\text{amp } z = \frac{5\pi}{6}$
 (C) $|z| = 2^{13} + \frac{1}{2^{13}}$ and $\text{amp } z = \frac{\pi}{6}$ (D) $|z| = 2^{12} + \frac{1}{2^{12}}$ and $\text{amp } z = \frac{7\pi}{6}$
18. $\sum_{n=9}^{575} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right) =$
- (A) $\frac{7}{24}$ (B) $\frac{23}{24}$ (C) $\frac{7}{8}$ (D) $\frac{11}{12}$
19. If S_n denotes sum of first n terms of a series and if $S_{n+2} - S_n = (n+2)^2$, then S_{10} is (given that $T_1 + T_2 = 4$)
- (A) 1540 (B) 220 (C) 1536 (D) 216
20. If z is a complex number satisfying the equation $z^6 + z^3 + 1 = 0$. If this equation has a root $e^{i\theta}$ with $\pi < \theta < \frac{3\pi}{2}$, then value of θ is:
- (A) 200° (B) 240° (C) 220° (D) 250°

SPACE FOR ROUGH WORK

SECTION-2

This Section contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99 (both inclusive).

1. The first term of a geometrical progression is equal to $p - 2$, the third term is $p + 6$ and the arithmetic mean of first and third term to the second term is in the ratio 5 : 3. Find positive integral value of p .
2. For a complex number z , if one root of the equation $z^2 - az + ib = 0$ is $1 + 2i$ and its other root is α , then the value of $\operatorname{Re}(\alpha) + |b|$ is _____ ($a, b, \in R$).
3. If a, b, c are distinct integers divisible by '3', then minimum value of $|a + b\omega + c\omega^2|^2$ is _____
4. If a, b, c, d are positive numbers such that $abcd = \frac{1}{6}$ and $\frac{a}{2} + \frac{b}{3} + \frac{c}{4} + \frac{d}{9} = \frac{2}{3}$, then $a + b + c + d$ is _____.
5. If $1^{-3} + 2^{-3} + 3^{-3} + \dots \infty = p$ and $1^{-3} + 3^{-3} + 5^{-3} + \dots \infty = q$, then $\frac{p}{q}$ is $\frac{a}{b}$
(where a, b are positive co-prime integers), then $a + b$ is:

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